

Unified Military Analytics and Comparison Dashboard – DV

Technical Documentation

This document provides an overview of the technical work carried out for KPI feature engineering and dashboard development as part of the Unified Military Analytics and Comparison Dashboard – DV project. The implementation focuses on structured data processing using Python and interactive visualization using Power BI, following industry-standard analytics practices.

1. Module 3 – KPI Feature Engineering

1.1 Objective

The objective of Module 3 is to transform cleaned military data into analytical, business-ready KPIs. This module establishes the foundation for all subsequent analysis and visualization.

1.2 Input Data

The input to this module is the dataset `military_cleaned.csv`, which contains standardized numerical values related to military assets, manpower, population, and economic indicators.

1.3 KPI Development

Multiple KPIs were engineered to capture different dimensions of military capability. Assets Per Capita was calculated to normalize total military assets by population size. Budget to GDP Ratio was computed to evaluate defense spending priority relative to economic capacity.

In addition, composite power indices were created for manpower, air power, land power, naval power, and economic strength. Each component was normalized to ensure fair comparison across countries.

1.4 Overall Power Index and Ranking

Normalized component indices were aggregated to form an Overall Power Index for each country. Countries were ranked using a dense ranking approach based on this index.

A Power Index Rank Gap metric was derived to represent the relative distance of each country from the top-ranked military power.

1.5 Metadata Enrichment

To support advanced filtering and segmentation, additional metadata attributes were added, including continent, region, and NATO membership flag.

1.6 Module 3 Output

The output of Module 3 is the file `military_final.xlsx`, containing two structured sheets. `Wide_Data` provides KPI values and metadata in a denormalized format, while `Long_Data` supports flexible analytical visualization.

2. Module 4 – Dashboard Planning and Prototyping

2.1 Objective

The objective of Module 4 is to convert engineered KPIs into an interactive dashboard that provides a high-level overview of global military analytics.

2.2 Data Integration

Power BI Desktop was used as the visualization platform. The `Wide_Data` sheet from `military_final.xlsx` was directly loaded into Power BI without applying transformations, ensuring a clean separation between data engineering and visualization layers.

2.3 Dashboard Design – Quick Stats

A single operational dashboard titled Quick Stats was developed for this phase. The dashboard is designed to provide immediate insight through summary KPIs and comparative visuals.

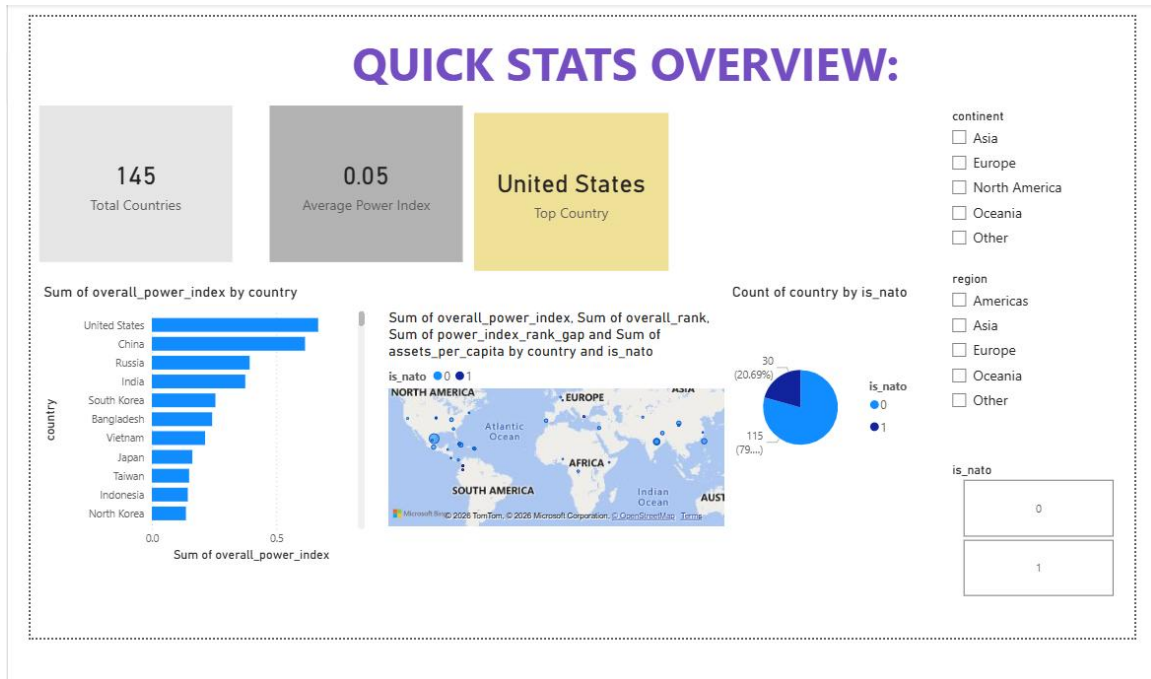
2.4 Visual Components

The dashboard includes KPI cards displaying total number of countries, average power index, and the top-ranked country. A bar chart highlights the top ten military powers based on overall power index.

A pie chart represents the distribution of NATO versus non-NATO countries. Interactive slicers for continent, region, and alliance membership enable dynamic filtering across all visuals.

2.5 Outcome

The Quick Stats dashboard validates correct KPI mapping and enables user-driven analysis. The design is scalable and can be extended with additional dashboards in future phases.



3. Conclusion

Modules 3 and 4 together establish a complete analytics workflow. Python-based KPI engineering ensures accurate and scalable data preparation, while Power BI visualization enables intuitive and interactive analysis. The implementation aligns with enterprise business intelligence standards and supports future expansion of the analytics platform.